

Estimating Nonlinear Insulin-Glucose Dynamics in Critically-ill Patients

Markus Mulvihill

KTH Royal Institute of Technology

August 7th, 2026



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Thesis Presentation

Estimating Nonlinear Insulin-Glucose Dynamics in Critically-ill Patients

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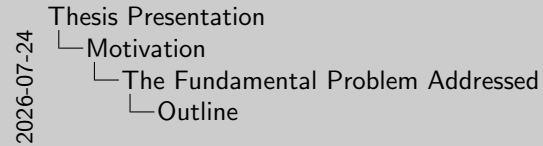
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1. Supervisor - Martin Lindström
2. Examiner - Ragnar Thobaben
3. Opponent - Divyayan Day

- 1 Motivation
 - The Fundamental Problem Addressed
 - Previous Work
- 2 Results and Contributions
 - Main Results
 - Key Ideas Behind Implementation
- 3 Summary
 - Takeaways
 - Further Reading



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Stress Hyperglycaemia in ICU Patients

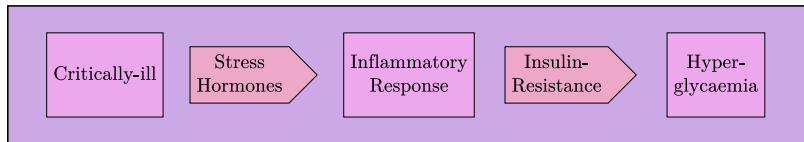
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Motivation

The Fundamental Problem Addressed

Stress Hyperglycaemia in ICU Patients



1. Inflammatory response secretes stress hormones
2. Acute myocardial infarction, sepsis, stroke

- **Problem:**

- **A solution:**

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└─ Motivation
 └─ The Fundamental Problem Addressed
 └─ Addressing Hyperglycaemia

Addressing Hyperglycaemia

● Problem: ● A solution:

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Stress-induced hyperglycaemia
increases the likelihood of
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Addressing Hyperglycaemia

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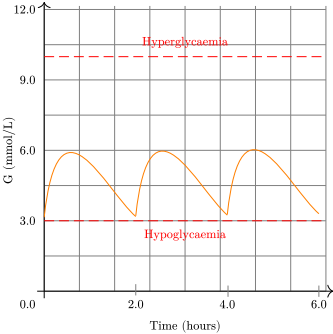
-Addressing Hyperglycaemia

Addressing Hyperglycaemia

- | | |
|---|--|
| <p>Problem:
Stress-induced hyperglycaemia increases the likelihood of mortality.</p> | <p>A solution:
Maintaining glycaemic levels to 4-6 mmol/L reduced mortalities by 50% (Berghe et al. 2001).</p> |
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- **Problem:**
Stress-induced hyperglycaemia **increases** the likelihood of **mortality**.
- **A solution:**
Maintaining glycaemic levels to 4-6 mmol/L reduced mortalities by 50% (Berghe et al. 2001).

Limitations of Intensive Insulin Therapy



- What:
- Why:

Figure: Tightly Controlled Blood Glucose

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- └ Motivation
- └ The Fundamental Problem Addressed
- └ Limitations of Intensive Insulin Therapy

Limitations of Intensive Insulin Therapy

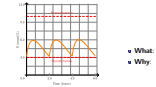


Figure: Tightly Controlled Blood Glucose

- What:
- Why:

Limitations of Intensive Insulin Therapy

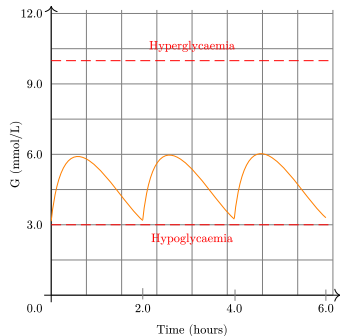


Figure: Tightly Controlled Blood Glucose

- **What:**
Poor patient adaptability results in cases of **hypoglycaemia** and **high glycaemic variation**.
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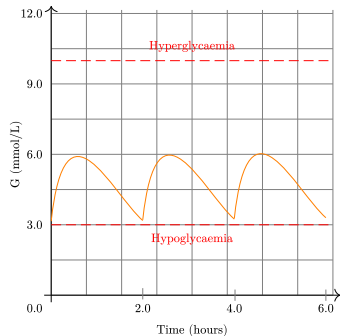


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Poor patient adaptability results in cases of **hypoglycaemia** and **high glycaemic variation**.
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Solely utilizing insulin infusions struggles to address patient variability.

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Limitations of Intensive Insulin Therapy



Figure: Tightly Controlled Blood Glucose

- **What:**
Poor patient adaptability results in cases of **hypoglycaemia** and **high glycaemic variation**.
- **Why:**
Solely utilizing insulin infusions struggles to address patient variability.

Infrequent measurements, not considering nutrition, and not complex to monitor dynamics changes in patient health

Role of Model-based Glycaemic Control

- Maybe do a nice photo showing how to convert insulin to equations instead of this long boring text
- Mathematical formulations of pancreas function provides more insight into glycaemic levels and endogenous insulin production

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- Consequently, physiological models better capture patients' insulin-glucose dynamics to sustain glycaemic ranges

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–Role of Model-based Glycaemic Control

1. Better method than IIT to be specific

- ◆ Maybe do a nice photo showing how to convert insulin to equations instead of this long boring text
- ◆ Mathematical formulations of pancreas function provides more insight into glycaemic levels and endogenous insulin production
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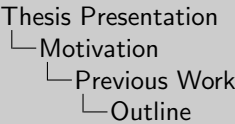
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Intensive Control Insulin-Nutrition-Glucose (ICING) model

Lin et al. has developed a clinically-validated insulin-glucose model with the following main components

Dynamic	Notation	Value
Blood Glucose	$\dot{G}$	mmol/L
Interstitial Plasma	$\dot{Q}$	mU/L
Plasma Insulin	$\dot{i}$	mU/L
Glucose in the Stomach	$\dot{P1}$	mmol
Glucose in the Gut	$\dot{P2}$	mmol
Endogenous Insulin Production	$u_{en}(t)$	mU/min
Insulin Sensitivity	$S_I(t)$	-
Input	Notation	Value
Exogenous Insulin	$u_{ex}(t)$	mU/min
External Food	$D(t)$	mmol/min
Parenteral Glucose	$P_N(t)$	mmol/min

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Computational Approximation of the ICING model

- The ICING model

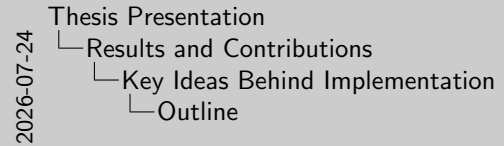
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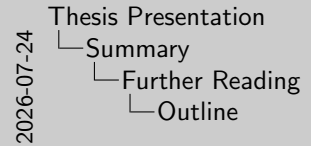
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 Berghe, Greet Van den et al. (Nov. 2001). “Intensive Insulin Therapy in Critically Ill Patients”. In: DOI: 10.1056/NEJMoa011300.

 Lin, Jessica et al. (May 2011). “A physiological Intensive Control Insulin-Nutrition-Glucose (ICING) model validated in critically ill patients”. In: DOI: 10.1016/j.cmpb.2010.12.008.

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-  Berghe, Greet Van den et al. (Nov. 2001). “Intensive Insulin Therapy in Critically Ill Patients”. In: DOI: 10.1056/NEJMoa011300
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